

FORENCHAIN

Blockchain-Based Chain of Custody Management System for Digital Forensics Compliance with Malaysian Forensic SOPs and Legal Standard.



UNIVERSITY OF
PLYMOUTH

TAARSINII PARTHEEPAN
BSCS2406001
BSC(Hons) Cyber Security)

Background & Problem Statement

- Digital evidence is easily altered or disputed
- Existing systems lack tamper-proof custody records
- Malaysian courts require strong evidence integrity

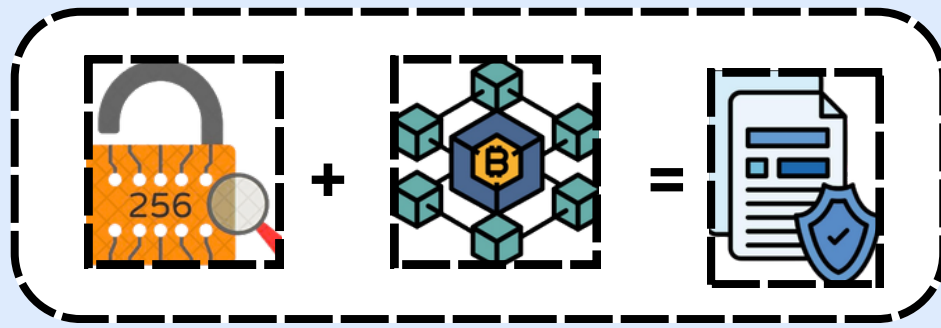
📌 **Reported in Malaysian Cybercrime & Forensic Reports (2023)**

Objectives

- Ensure tamper-proof management of digital evidence.
- Provide verified chain of custody records.
- Improve legal admissibility in Malaysian courts.
- Enhance trust and accountability in forensic workflows.

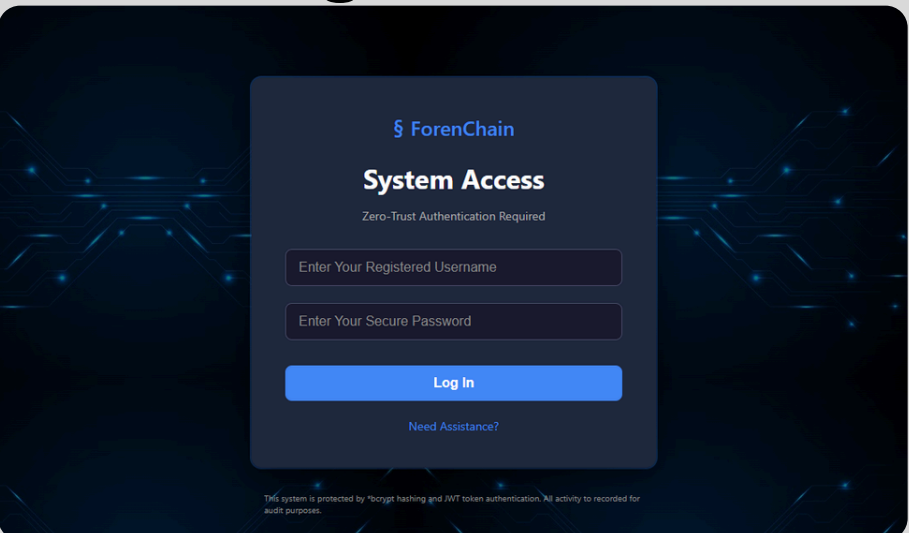
Core Features

- 🔒 **SHA-256** cryptographic hashing
- 🔗 **Hash chaining** (linked custody records)
- 🔗 **Blockchain anchoring** (Eth)
- 👤 **Role-based access control**
- 📄 **Secure audit logging**



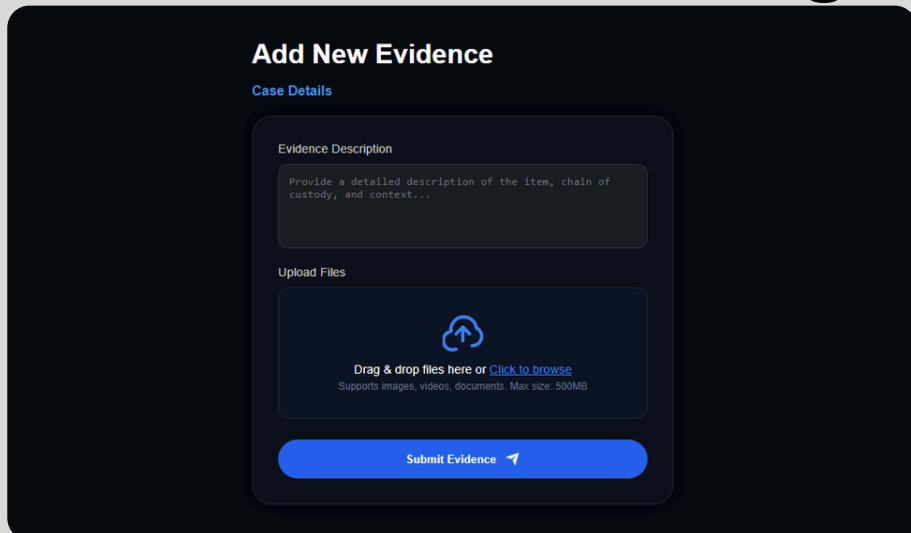
System Flow

Secure Login & Role Control



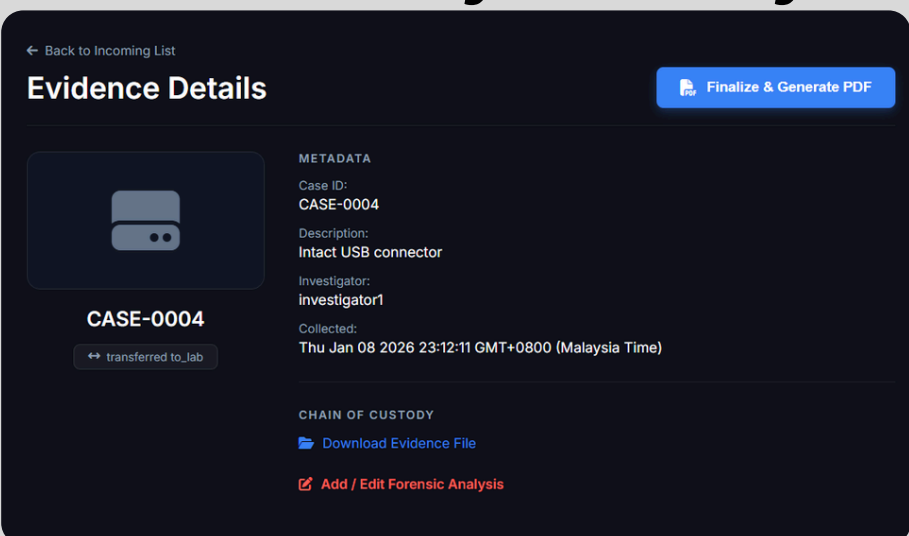
Only authorised users can access the system. Each user has a specific role and responsibility.

Evidence Collection (Investigator)



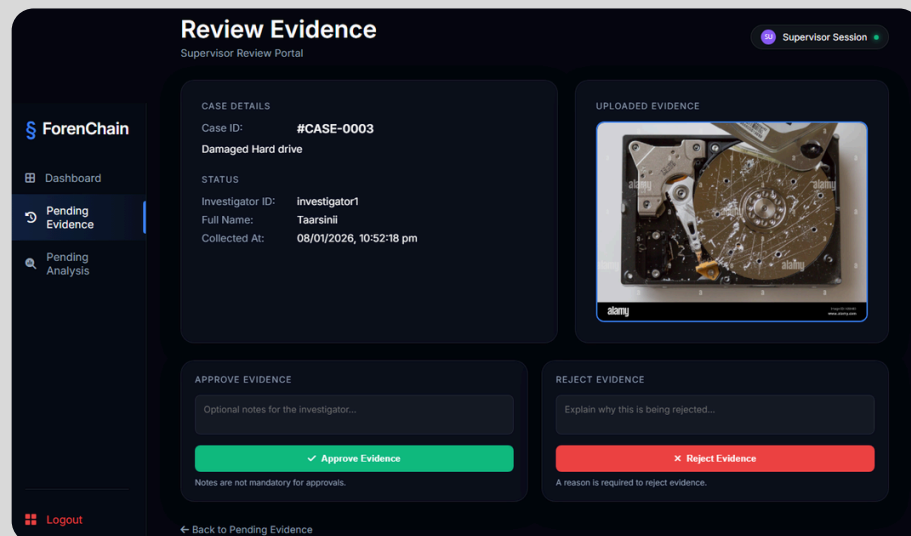
investigator uploads digital evidence. A unique digital fingerprint (SHA-256 hash) is created immediately.

Forensic Analysis (Analyst)



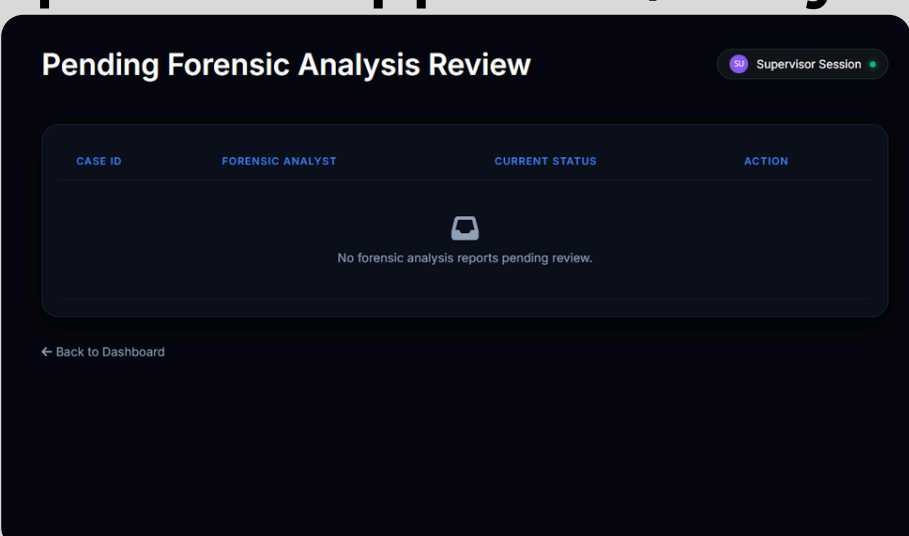
Examines the evidence and prepares a forensic report without modifying the original evidence.

Supervisor Approval (Evidence)



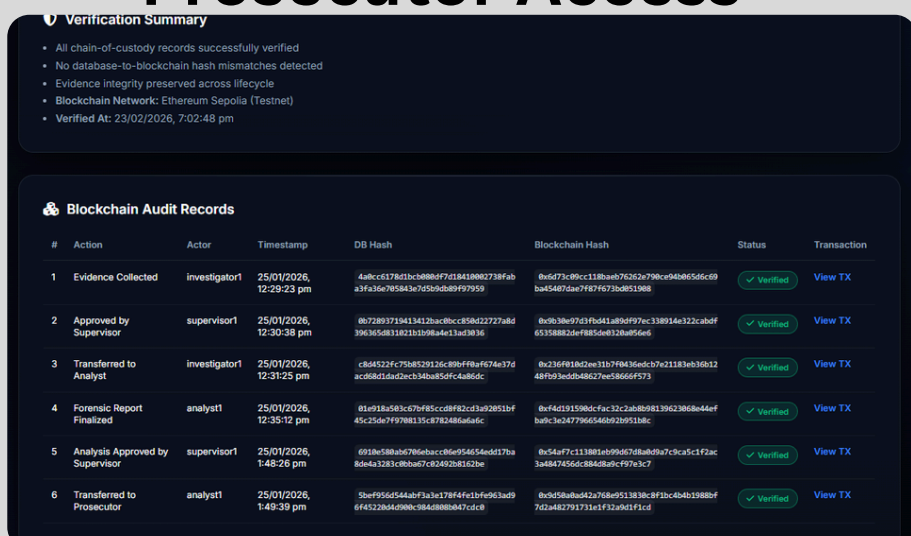
supervisor reviews and approves the collected evidence before it can proceed

Supervisor Approval (Analysis)



Verify the Forensic analysis + attach CCTV footage taken during the analysis

Blockchain Anchoring & Prosecutor Access



The full evidence history is permanently recorded and can be reviewed by prosecutors

Security Architecture

Security Architecture = SHA-256 + Blockchain

SHA-256 Hash Chaining: **Blockchain Anchoring:**

- Links every custody event
- Detects internal tampering
- Ensures continuity
- Immutable public verification
- Prevents insider manipulation
- Court-admissible proof

How ForenChain aligns with CoC?

- **WHAT:** Unique evidence ID & hash
- **WHO:** Role-based user tracking
- **WHEN:** Timestamps + blockchain
- **WHERE:** Controlled system storage
- **WHAT HAPPENED:** Hash-chained custody log
- **INTEGRITY:** Hashing + blockchain proof

Future Development

- 🔒 **Evidence encryption** (AES-256)
- ✍️ **Digital Signatures**
- 📱 **Multi-Factor Authentication** (Beyond OTP)
- 🚨 **Automated integrity alerts**

